Unit: Unit 3: Functions **Name:** ____________
Topic: Definition of Relation and Function, Domain, Range, and Notations **Date:** ________

Foundation (Jalapeno)

Q1. Which of the following is a function?
(A) $y = 2x + 3$
(B) $x^2 + y^2 = 4$
(C) $y = \pm\sqrt{x}$
(D) $x = y^2$
(E) $y = |x|$

Q2. What is the domain of the function $f(x) = \frac{1}{x}$?
(A) $(-\infty, -1) \cup (1, \infty)$
(B) $(-\infty, 0) \cup (0, \infty)$
(C) $(-\infty, 0)$
(D) $[0, \infty)$
(E) $(-\infty, \infty)$

Q3. A basketball player scores $f(x) = 2x + 5$ points in x games. What is $f(3)$?
(A) 6
(B) 11
(C) 15
(D) 20
(E) 25

Q4. Which of the following relations is a function?
(A) $\{(1, 2), (2, 3), (3, 2)\}$
(B) $\{(1, 2), (1, 3), (2, 3)\}$
(C) $\{(1, 1), (2, 2), (3, 3)\}$
(D) $\{(1, 2), (2, 2), (3, 2)\}$
(E) $\{(1, 1), (2, 1), (3, 1)\}$

Q5. What is the range of the function $g(x) = x^2 - 4$?
(A) $(-\infty, -4]$
(B) $[-4, \infty)$
(C) $(-\infty, 0)$
(D) $[0, \infty)$
(E) $(-4, 4)$

Q6. A tennis player runs $h(x) = 3x - 2$ miles in x hours. What is $h(2)$?
(A) 2
(B) 4
(C) 6
(D) 8
(E) 10

Q7. What is the domain of the function $f(x) = \sqrt{x - 2}$?
(A) $(-\infty, -2)$
(B) $(-2, 2)$
(C) $[2, \infty)$
(D) $(-\infty, 2)$
(E) $(-2, \infty)$

Q8. Which of the following statements is true about the function $y = |x|$?
(A) It is an odd function
(B) It is an even function
(C) It is a one-to-one function
(D) It is a decreasing function
(E) It is a constant function

Open Ended Questions (FRQ)

Q9. Determine the domain and range of the function $f(x) = \frac{x+1}{x-1}$. Show your work and explain your reasoning.

Q10. A cyclist travels $d(x) = 2x^2 + 5x - 3$ miles in x hours. Evaluate $d(4)$ and explain what this means in the context of the problem. Show your work.

Chef's Tips

- Provide clear instructions for each question
- Ensure students have the necessary materials and equipment
- Allow students to ask questions and seek help when needed